

- *Rule 5.* When looking at the quantity $\sum_i (X_i - \bar{X})^2$ it can be deduced that distributing calibrators at the ends of the calibration range is more valuable since it is maximized. One of the well-known properties of least-squares technique is that better estimations of the regression coefficients are obtained when two levels of concentration are set far apart in the calibration interval. For instance, international guidelines to analytical validation, such as FDA and EMA recommend performing a calibration with at least six calibrators, two of them located at expected upper and lower limits of quantification (ULOQ and LLOQ) in the calibration design. However, no other calibrator locations are indicated.

Official guidelines have usually recommended most of the rules listed above. This leads to satisfactory inverse-prediction when dealing with External Calibration (EC) in the linear case. While only a few formal expectations are provided for other types or models of analytical calibration and the multipoint matrix-matched external calibration (MMEC) is recommended by international guidelines for the validation of bioanalytical methods [9]. This mention of official guidelines suggests a growing interest in the need for new analytical calibration practices. Currently, the possibility of cross validating the results obtained with an alternative analytical method to a reference one allows the analyst to investigate alternative quantification methodologies without sacrificing performance.

“Obligations of result” and “obligations of means” have a longstanding history in private law. The obligation of result is simply the obligation of the debtor to attain a predetermined result. These legal concepts could be transposed to the organization of the method validation. For instance, official recommendations are often based on an “obligation of means” such as a minimal number of calibrators or replicates for calibration conditions. If an “obligation of results” were prescribed, such as participating in systematic proficiency testing (Section 4.3), the analyst would be able to select the calibration organization most appropriate to the laboratory.

The implementation of this other approach may promote renewed criteria to evaluate the calibration performance, including a focus on the observed results and their respective uncertainties. The comprehensive validation approach is probably the better way to establish innovative quantification methodologies for modern instrumentation. It also means that calibration is a fully integrated part of the analytical process.

2.5.2 Calibration as a Source of Uncertainty

When considering MU estimation of a quantitative result, the calibration step remains one of the key stages to reliably assess its quantification. Unfortunately, it seems impossible to estimate the specific role of calibration in MU. Only a comparative approach is applicable using different calibration models or algorithms. Since the 1990s, metrologists have developed what is called the Uncertainty Approach in opposition to the more traditional Error Approach. To estimate the MU of any reported measurement value, an effective solution consists of applying the

uncertainty approach and considering calibration as one of the many sources of uncertainty participating in the combined MU of an analytical result, noted $u_c(Z)$ with subscript c stands for combined.

As recalled above, many studies have investigated the influence of different experimental designs and strategies applicable to calibration, such as WLS regression, replication of measurements or optimized distribution of calibrators within the calibration interval. If they are beneficial, they should be evaluated due to the reduction of $u_c(Z)$.

The estimation of MU in analytical sciences is still at its beginnings, and no absolute procedure is recognized by analysts. Examples applicable to a specific operating procedure or field of analysis have been published. However, the role of calibration in MU must be scrutinized. The general procedure to estimate MU, is described in Chapter 6. It is relatively straightforward and consists of four steps: specify the measurand; identify the uncertainty sources; simplify and quantify the uncertainty components; and calculate the combined uncertainty. Calibration is one of the many sources of uncertainty in analytical sciences which can be identified in the second step. Whatever the calibration procedure, at least three major sources are present:

- The chosen calibration model.
- Applied algorithm to estimate model coefficients.
- In some cases, calibrant uncertainty in the reference material used for calibrator can also be accounted for.

However, it is difficult to define an experimental design to separately estimate the uncertainty linked to calibration. Therefore, the role of calibration in MU can only be indirectly estimated as already underlined. In some statistical textbooks, strategies to select the *best* calibration model and minimize the importance of calibration as an uncertainty source are proposed. Unfortunately, in analytical sciences the X concentration and Y response are naturally highly correlated and classical statistical criteria, such as the r^2 or AIC , are inefficient for selecting the most appropriate calibration model. Several other techniques exist, such as the visual inspection of residuals or dedicated statistical testing to verify the lack-of-fit or the homoscedasticity.

But experience shows that all these techniques are not very efficient because of the extremely strong causal relationship between X and Y . The Guide to the expression of Uncertainty in Measurement (GUM) general procedure recommends regrouping sources of uncertainty, and a more empirical approach consists of making several estimations of the combined MU using different calibration models and regression techniques. For example, using QC values or the method accuracy profile are possible alternatives to reach this goal. To obtain reliable quantification in terms of trueness and precision, the analytical calibration function must take the response relationships for all relevant analytes and interferences into account. From a practical perspective, two primary prerequisites must be fulfilled:

- Physicochemical properties of surrogate standard must be as close as possible to the authentic analyte in terms of solvent and nature.

- Calibration standards and working samples must have comparable behavior in the measurement system.

Finally, many MU guidelines explain how to estimate MU for one working sample. In many contract laboratories, the composition of received sample may be highly variable. The application of the proposed procedures to each working sample would be tedious and difficult. In Section 7.6.3 we promote the concept of Uncertainty Function which represents another approach more suitable to analyst's daily practice.

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3

Precision

3.1 Outputs of Interlaboratory Studies

3.1.1 Diverse Precision Parameters

Precision is one of the major characteristics of any measurement method and is defined in clause 2.5 of the VIM [1]:

Precision is the “closeness of agreement between measured quantity values obtained by replicate measurements on the same or similar objects under specified conditions.”

The definition of precision conveys the idea that replicate measurements, when made on the same sample with the same method, are dispersed in a specific way. An early challenge for the official control bodies, which needed to compare data of different origins, was to understand how precision may vary from one laboratory to another and how they could rely on analytical data to monitor different tasks such as people's health or environmental pollution. While trade became more global, the concern of economic agents was similar for product exchanges. It was concluded that it was essential to have a common procedure for comparing measurement values obtained by different laboratories and to ascertain if the discrepancies were caused by the method or the laboratory.

Therefore, it was decided to assess precision within the framework of interlaboratory studies. Early studies were organized in the 1940s. In the 1990s, ISO published a set of five international standards under the global reference ISO 5725 [2–6]. Several parts of this major standard are under revision. The general scope was to define a harmonized procedure for the statistical interpretation of interlaboratory studies and the computation of precision parameters. The principal role was to underline that the estimation of precision depends on the condition where replicates are conducted, i.e. which sources of variations are accounted for:

- When replicates are made in the same laboratory without modifying anything between two measurements, only random effects are counted, and the precision is estimated under *repeatability condition*.